Fiscal Year:	FY 2005	**Task Last Updated:**	FY 02/02/2005
PI Name:	Thompson Jr, James N. Ph.D.		
Project Title:	Genetic and Developmental Stability in Response to Long-Term Exposure of Drosophila melanogaster to a Space Station Environment		
Division Name:	Space Biology		
Program/Discipline:	SPACE BIOLOGY		
Program/Discipline--Element/Subdiscipline:	SPACE BIOLOGY--Organismal biology		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	(1) Cell & Molecular Biology (2) Animal Biology: Invertebrate		
Space Biology Cross-Element Discipline:	(1) Developmental Biology		
Space Biology Special Category:	None		
PI Email:	jthompson@ou.edu	**Fax:**	FY 405-325-6202
PI Organization Type:	UNIVERSITY	**Phone:**	405-325-4821
Organization Name:	University of Oklahoma		
PI Address 1:	Department of Zoology		
PI Address 2:	RH 314		
PI Web Page:	http://faculty-staff.ou.edu/		
City:	Norman	**State:**	OK
Zip Code:	73019-3032	**Congressional District:**	4
Comments:			
Project Type:	Flight	**Solicitation / Funding Source:**	2001 Space Biology ILSRA
Start Date:	08/01/2002	**End Date:**	09/30/2007
No. of Post Docs:	0	**No. of PhD Degrees:**	0
No. of PhD Candidates:	2	**No. of Master' Degrees:**	0
No. of Master's Candidates:	2	**No. of Bachelor's Degrees:**	2
No. of Bachelor's Candidates:	4	**Monitoring Center:**	NASA ARC
Contact Monitor:		**Contact Phone:**	
Contact Email:			
Flight Program:	ISS		
Flight Assignment:	Currently in Phase B for feasibility and strain characterization studies. NOTE: NCE to 9/30/2007, per A Chu 7/2006.		
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Woodruff, Ronny Clifford (Bowling Green State University)		
Grant/Contract No.:	NCC2-1355		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	Using a simple invertebrate model, the fruit fly Drosophila melanogaster, we will study the effect of the space environment on: (1) successful reproduction over a period of seven generations on the International Space Station (ISS) in microgravity and 1 g control conditions; (2) genetic stability as measured by mutation accumulation during this multi-generation period; (3) the effect of microgravity, radiation, and other elements of the space environment on predicted reductions in developmental homeostasis (regulatory buffering) in response to environmental and/or mutational stresses; and (4) protein, DNA mutation, or RNA profiles from which stress mechanisms can be analyzed.. Mutation accumulation will be measured in genetic loci that cause an altered lifespan, that reduce viability (i.e., reduce relative fitness), that contribute to increased heritable variation in neurosensory organ number (principally sternopleural bristle number), and that yield DNA microsatellite and sequence variation. Changes in expression of genetic repair and other stress-related genes by proteomic profiles, RNA microarray, or comparable techniques is also of interest. The primary morphological characters assayed for developmental stability will be changes in the pattern of wing support vein patterns and changes in the number and distribution of surface sensory organs (e.g., bristles and the campaniform sensillae located on the third longitudinal [L3] vein). A deviation from symmetry (fluctuating asymmetry, FA) measures the quality of developmental regulation in traits that are normally symmetrical. Together, these measures of genetic and developmental stability will extend our understanding of the rate and types of genetic changes that can occur during long-term exposure to a space environment and the effects of this novel stress on developmental homeostasis. This helps answer key questions about the ability of biological systems common to all organisms to adapt to novel environmental conditions of a space environment. The scientific goals for the second year (Phase B) focus on the characterization of breeding programs and prospective strains of Drosophila melanogaster for planned implementation of a multi-generation mutation accumulation experiment on the International Space Station. While the hardware is being developed for this ISS experiment, the goals are to: (1) characterize selected strains for mutation sensitivity, sequence DNA at targeted sites, carry out microsatellite allele profiles, and measure baseline fluctuating asymmetry (FA) of morphological markers; (2) initiate RNA stability tests and evaluate alternative methods for maintaining stable DNA, RNA and protein samples under conditions available on the ISS; (3) characterize the Binscy Assay for mutation accumulation over multiple generations as a basis for later evaluation of biocompatibility tests in the ISS habitats; and (4) initiate tests of videoimaging quality to assess diagnostic morphological traits during generations on the ISS.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	The results of a long-term exposure to the radiation and gravitational condiditions of a space environment will extend our knowledge of the sensitivity and responsiveness of genetic and developmental systems to novel stresses. This information will help assess measures required for successful colonization and utilization of space.
	The experimental goals for Year 2 (Project Phase B) have been achieved with great success. Discussions about the design of Drosophila habitat chambers in the European Modular Cultivation System (EMCS) were held at Ames Research Center in July and October 2004. At these times, some hypergravity and vibration experiments were also conducted to characterize further the prospective strains to be used in the ISS experiments. A site visit to the University of Oklahoma by four EMCS scientists and engineers occurred in August 2004. Further discussions were also held at conferences in Washington, DC, in March and New York in November 2004. Following these interactions, a revised Experiments Requirement Document for the International Space Station flight experiment was prepared and submitted by Araceli Maldonado, Lockheed Martin Space Operations, NASA Ames Research Center. A key experiment planned for the International Space Station is a multi-generation mutation accumulation assay, the “Binscy Assay”, designed to allow new mutations to collect on a balanced (i.e., stably heterozygous) X chromosome. This year we assessed our ability to visualize sex and eye genotype (the Bar eye is narrow, marking chromosomes on which mutations accumulate; the “control” chromosome marker males have round red eyes), using the camera system for the culture chamber being designed by the EMCS group at NASA Ames Research Center. A mixture of males and females from the Binscy Assay were placed in a culture chamber and videotaped using a computer-controlled remote camera comparable to ones being tested for the ISS habitat. At this stage the principal questions were (1) can one discriminate males versus females; (2) can one discriminate the narrow Bar eye phenotype of Binscy males in contrast to the round eye of C(1;YS)oc ptg males; and (3) are still images sufficient to assess traits, or are video sequences necessary? The experiments were successful in answering these questions. Even though the camera system has not yet been optimized for the EMCS Drosophila habitat under development now, the magnification and depth of focus are sufficient to determine sex and eye phenotypes. But while still images can allow us to assess culture status, the videoimages are required for phenotype assays since the flies are in motion through the focal plane. By moving through this plane, the flies essentially focus themselves from time to time in the field of vision. Mutation accumulation experiments with Binscy confirm it is an effective breeding program for the planned ISS experiments. In three replicate multiple-generation experiments, the mean viability went up for each. Using the Bateman-Mukai method and considering that the X chromosome contains about 10% of the total genes in this species, our average estimated genomic mutation rate, U, is 0.45 and the average effect of a deleterious mutation on viability is 0.14. Our estimated U is about in the middle of the previously reported U values. The use of mutation accumulation and control siblings raised together makes this a robust experimental design that overcomes many of the criticisms brought against other approaches to measuring U. It proves itself to be an excellent choice for an ISS experiment on genomic mutation. We are also determining microsatellite allele genotypes of the Binscy X chromosome and of Canton-S, a commonly-used wild type (normal) strain. Microsatellite alleles are assayed by isolating DNA from single flies, amplifying a specific microsatellite locus by PCR, and separating bands using Spreadex gels (Elchrom) on the SEA 2000 electrophoresis system. This system has the resolution to separate PCR fragments that differ by only a few nucleotides. We are now determining which specific microsatellite loci can be run together in the same gel lanes to optimize the data generated per gel. Multiplexing the PCR reactions is also going well. Proteomics techniques based on the recently achieved genome databases and developments in modern mass spectrometry can identify proteins and differentiate the levels of protein expression between animals in normal and stressed states. This year we have worked with new collaborators to begin assessing the potential to use a proteomics approach as a complement to the originally planned DNA and RNA assays from ISS samples. We now have the experimental setup

Task Progress:	and expertise (1) to run 2-D gels to display proteins of control and experimental animals, (2) to quantify the amount of each protein by scanning and integrating image density, and (3) to identify the proteins that changed expression by stress. Protocol tests are being done to determine the best preservation with various stabilizing solutions, some of which may be compatible with sampling DNA and/or RNA at the same time. For example, following 15 minutes of vibration (20 to 2000 Hz; Ames Research Center), we observed that the treated females were inactive and remained sluggish for the next hour until flies were placed in liquid nitrogen to begin protein analysis. Vibrated males and control flies, on the other hand, behaved normally. Protein spots designated 1, 2, 3, 4, 5, and 6 in vibrated females (PVF; figure available on request) do not exist in the male gels. Spots 1, 2, and 3 in PVF do not exist, or are greatly reduced, in control females. The protein was subsequently identified as Yolk Protein 1 and the posttranslational modification events are now being characterized further. Such assays can help us understand the molecular and physiological events associated with stress exposures like vibration. Proteomic profiles of other key tissues like ovaries and other stresses like elevated background radiation are also underway. This promises to be a powerful and sensitive tool for assessing both posttranslational protein modifications and changes in gene activity associated with stress and adapation to the ISS space environment. RNA stability tests show that commercial products like RNAlater (Sigma Chemical Company) are suitable, as long as a sampling tool is designed to break the adult cuticle so solution can penetrate tissue quickly. Protein samples depend upon rapid dehydration, and ethanol buffer solutions are satisfactory. Acetone is also suitable, but safety concerns are being studied. Similarly, ethanol-based buffers have been shown to maintain morphological characters well for microscopic analysis of fluctuating asymmetry even months later. We now routinely microdissect wings from adults stored in 70% ethanol + 5% glycerol solutions for 6 months or more, often including disturbance from shipment from the Woodruff lab in Ohio to the Thompson lab in Oklahoma. But substitutes for ethanol-based buffers may be required. Thus, we are also comparing the success of molecular and morphological storage using other alcohols and new commercial products. Fluctuating asymmetry (FA) is a measure of the difference between the expression of a symmetrical character on the left versus right sides of an individual. When measured with appropriate replications and experimental design, it is an effective measure of developmental stability (homeostasis). Our primary traits are pattern elements in the Drosophila wing. In addition to characterizing Canton-S under control, hypergravity, vibration, and radiation stress conditions, we have tested and confirmed our ability to microdissect intact wings easily from ethanol-glycerol and other long-term storage conditions. Furthermore, an hormesis-like response (benefit of low-level stress in protecting against later and stronger stress exposures) has been observed in our experiments on the effects of vibration. The “heat shock response” and function of proteins like Hsp90 may provide a mechanistic explanation. We are pursuing this using both morphometric and proteomic assays to define more precisely the molecular and developmental predictions we should make about brief versus long-term exposure to space environment stresses on the ISS. Finally, we have confirmed that some traits may be more sensitive than others as a measure of developmental instability. The wing landmarks are measured on a continuous quantitative scale and appear to be quite appropriate to the assays we have designed for the ISS experiment. Others, like the metric counting of sensory bristles, are limited by varying in presence/absence steps. Initial evaluation of key generations shows a small increase in wing morphology FA in replicates accumulating mutations due to elevated background radiation. Data from metric measures of sternopleural bristle number, however, do not show a significant difference over generations in the Binscy Assay.

Bibliography Type:	Description: (Last Updated: 08/08/2019)
Dissertations and Theses	Madrid, D.J. "The effects of hypergravity and vibration on Drosophila melanogaster." Honors Thesis, Department of Zoology, University of Oklahoma, Norman, OK. May-2004
Dissertations and Theses	Tandy, M.L. "Somatic mutation and developmental stability in Drosophila." Honors Thesis, Department of Zoology, University of Oklahoma, Norman, OK. May-2004
Presentation	Gong, Y.; Thompson, jr., J.N.; Woodruff, R.C. "Estimation of deleterious genomic mutation rate of Drosophila melanogaster by the Binscy Assay." Society for the Study of Evolution Conference, 26-30 June 2004, Colorado State University, Ft. Collins, CO. Jun-2004
Presentation	Gong, Y.; Thompson, jr., J.N.; Woodruff, R.C. "Estimation of deleterious genomic mutation rate of Drosophila melanogaster by the Binscy Assay." Genomes and Evolution Conference, 17-20 June 2004, Penn State University, University Park, PA. Jun-2004
Presentation	Thompson, jr., J.N. "Flies in space: The challenge of adapting to a truly novel environment." Public Seminar, Department of Zoology, University of Oklahoma, Norman, OK Feb-2004
Presentation	Thompson, jr., J.N.; Woodruff, R.C. "Exploring the limits to adaptability: Drosophila as a model in hypergravity and vibration stress." Public Seminar, NASA Ames Research Center, Moffett Field, CA. Oct-2004
Presentation	Thompson, jr., J.N.; Woodruff, R.C. "A multi-generation mutation accumulation experiment for the International Space Station." “Flies in Space Workshop“, 45th Annual Drosophila Genetics Research Conference, 24-28 March, Washington, D.C. Mar-2004